

Heart Murmur Classification from PCG Recordings Using MFCC and Log-Mel Features: A Comparison with Deep Learning Approaches

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Abstract—This project investigates the use of Mel-frequency cepstral coefficients (MFCC) and log-Mel spectrogram features for classifying cardiovascular health from phonocardiogram (PCG) signals. Using the CirCor DigiScope dataset [1], which contains pediatric heart sound recordings from multiple auscultation sites, we applied signal segmentation, handcrafted feature extraction, unsupervised clustering, and supervised classification using K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Decision Trees. We evaluated classifier performance and interpretability, and compared our results against the deep learning approach proposed by Zhang et al. [2], which relies on parallel-attentive modeling and uncertainty estimation.

I. INTRODUCTION

Cardiovascular disease (CVD) remains a leading global health concern and is a primary cause of mortality worldwide. Early diagnosis through non-invasive techniques, such as cardiac auscultation, plays a vital role in detecting pathological heart sounds associated with valvular or structural abnormalities.

Phonocardiograms (PCGs), which record heart sounds via digital stethoscopes, provide valuable acoustic signals that can reveal the presence of murmurs, extra heart sounds, or rhythm irregularities. However, interpreting PCGs manually requires clinical expertise and can be subject to variability.

In this work, we explore the use of classical machine learning methods to classify heart sounds from PCG recordings using interpretable audio features. Specifically, we extract Mel-frequency cepstral coefficients (MFCCs) and log-Mel spectrogram features to represent the audio signals. We evaluate the effectiveness of these features using K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Decision Trees, with a focus on both classification accuracy and model interpretability.

II. DATASET

We used the CirCor DigiScope dataset [1], a publicly available collection of pediatric phonocardiogram (PCG) recordings collected from multiple auscultation locations: aortic valve (AV), pulmonary valve (PV), tricuspid valve (TV), mitral valve (MV), and an additional precordial site (Phc). The dataset was developed for the George B. Moody PhysioNet

Challenge 2022 and includes detailed annotations for heart murmur detection.

Each subject record includes:

- `.wav` files containing raw heart sound recordings for each auscultation site.
- `.hea` files providing WFDB header metadata.
- `.tsv` segmentation files that mark the timing of S1 and S2 heart sounds within each recording.
- `.txt` files with subject-level metadata, including age, sex, height, weight, pregnancy status (for female patients), and murmur characteristics.

File naming follows the convention `ABCDE_XY.wav`, where `ABCDE` is the anonymized subject identifier and `XY` denotes the auscultation location code.

III. METHODS

A. Segmentation and Labeling

Each phonocardiogram (PCG) recording was segmented into overlapping windows of 5 seconds with a 1-second hop size. This segmentation approach ensured sufficient temporal context for feature extraction while generating a larger number of training samples from each recording. The 5-second window length was chosen to match the duration of the shortest recording in the dataset, ensuring that all recordings could be fully processed without truncation or padding.

Labels were assigned at the subject level based on the associated metadata files. Specifically, we used the `.txt` file corresponding to each subject to determine murmur presence, and we used `.wav` recordings to extract the segments. The subject was labeled:

- `Present` — if any of the subject's recordings were annotated with a murmur.
- `Absent` — if no murmurs were detected and all annotations were definitive.
- `Unknown` — if murmur presence was marked as uncertain in any of the subject's recordings.

This subject-level labeling ensured consistency and avoided ambiguous training signals from differently labeled segments belonging to the same individual.

B. Feature Extraction

We extracted two types of audio features from each 5-second segment: Mel-frequency cepstral coefficients (MFCCs) and log-Mel spectrograms, both commonly used in speech and audio analysis for their ability to capture perceptually relevant frequency patterns.

Mel-frequency cepstral coefficients (MFCCs) model the short-term spectral envelope of a sound in a compact form inspired by the human auditory system. We computed 13 MFCCs per segment using the `librosa.feature.mfcc()` function. The process involved applying a discrete cosine transform (DCT) to the log-scaled Mel spectrogram, resulting in coefficients that summarize timbral properties of heart sounds.

Log-Mel spectrograms represent the distribution of signal energy over time across perceptually spaced frequency bands. Each spectrogram was computed using `librosa.feature.melspectrogram()` followed by logarithmic (decibel) scaling. We used 128 Mel bands and averaged the resulting time-frequency representation over the time axis to obtain a 128-dimensional feature vector for each segment.

All extracted feature vectors were stored in `.npy` files and indexed alongside filenames, patient IDs, segment indices, and labels in a centralized CSV file to support traceability and reproducibility.

IV. UNSUPERVISED ANALYSIS

To explore whether the extracted features revealed meaningful patterns, we applied Principal Component Analysis (PCA) to reduce the MFCC and log-Mel feature dimensions to 2D. We then used K-Means clustering with $k = 3$, a simple unsupervised algorithm that partitions data into k groups by minimizing intra-cluster variance.

A. Clustering Results

As shown in Fig. 1, the resulting clusters showed some local structure but did not align well with the true diagnostic labels—especially for the `Present` and `Unknown` classes. This highlights the limited effectiveness of unsupervised clustering in this context and supports the need for supervised classification.

V. CLASSIFICATION EXPERIMENTS

A. Data Splitting

To train and evaluate our models fairly, we split the dataset into 70% training, 15% validation, and 15% test sets. We used subject-level separation, meaning all recordings and segments from each subject were placed in only one of the splits. This prevents data leakage because heart sound patterns from the same individual do not appear in both training and evaluation sets.

B. Evaluated Models

We tested three standard classification algorithms using the `scikit-learn` library: `KNeighborsClassifier`, `SVC`, and `DecisionTreeClassifier`. Each model was selected for its interpretability or effectiveness in handling small to medium-sized feature sets.

- **K-Nearest Neighbors (KNN)**: A non-parametric, instance-based learning algorithm that classifies a data point based on the most common label among its k nearest neighbors in the training set.
- **Support Vector Machine (SVM)**: A supervised learning model that finds the optimal hyperplane to separate classes by maximizing the margin between them. We used the Radial Basis Function (RBF) kernel to allow for non-linear separation.
- **Decision Tree**: A tree-structured classifier that makes decisions by splitting features into subsets based on value thresholds. It is easy to interpret and useful for analyzing feature importance, though generally less accurate than more complex models.

C. Metrics

We evaluated model performance using the following metrics:

- **Accuracy**: The ratio of correct predictions to the total number of predictions. While easy to understand, it may be misleading for imbalanced classes.
- **F1-score**: The harmonic mean of precision and recall. It is more informative than accuracy when class sizes are uneven and is especially useful for measuring how well the model detects murmurs.
- **Macro-F1**: The average F1-score across all classes, giving equal weight to each class regardless of its frequency. It is useful for understanding performance on minority classes.
- **Confusion Matrix**: A table that shows how many examples from each true class were predicted as each possible class. It helps visualize errors like false positives and false negatives.
- **Weighted Accuracy**: A clinically weighted metric proposed in the PhysioNet 2022 Challenge [3]. It gives more importance to correctly classifying murmurs than normal cases. It is calculated as:

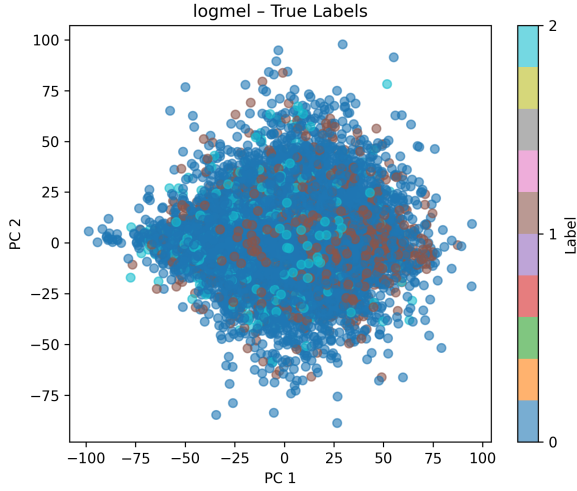
$$\text{Weighted Accuracy} = \frac{5 \cdot \text{TP}_{\text{Present}} + 3 \cdot \text{TP}_{\text{Unknown}} + 1 \cdot \text{TP}_{\text{Absent}}}{5 \cdot N_{\text{Present}} + 3 \cdot N_{\text{Unknown}} + 1 \cdot N_{\text{Absent}}}$$

where TP_{class} is the number of correct predictions for a class, and N_{class} is the number of samples in that class.

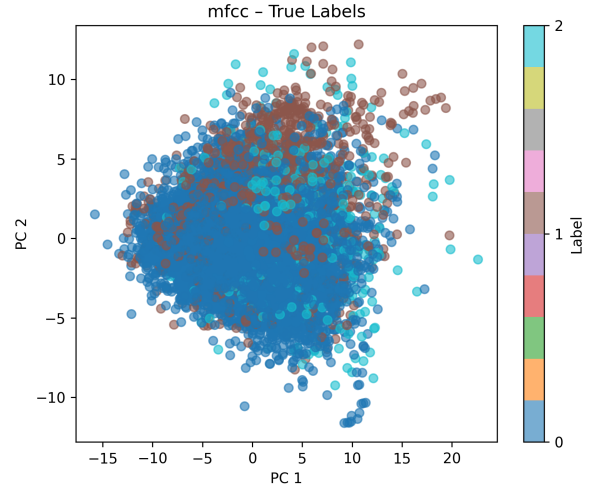
We used these metrics to match the official evaluation protocol of the Zhang et al. [2].

VI. RESULTS

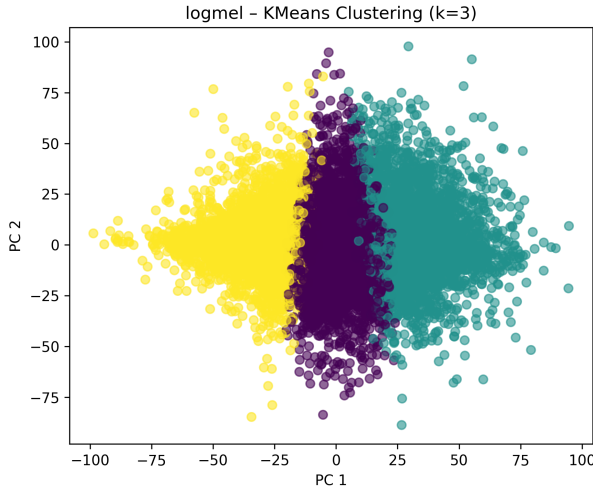
We evaluated all models on the test set using both MFCC and log-Mel features. Table I summarizes the key performance metrics: accuracy, F1-score, and weighted accuracy.



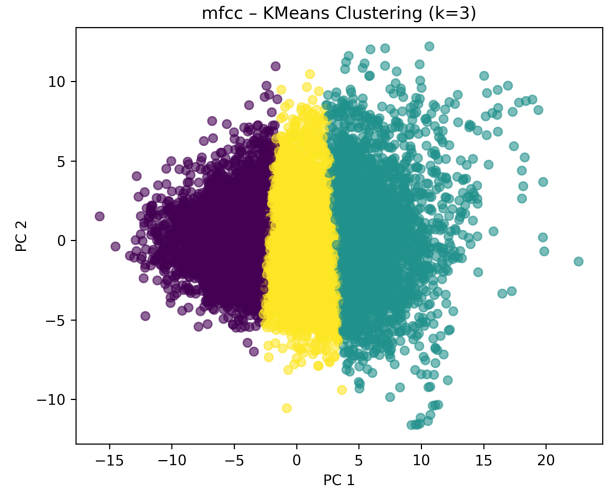
(a) Log-Mel: True Labels



(b) MFCC: True Labels



(c) Log-Mel: K-Means ($k = 3$)



(d) MFCC: K-Means ($k = 3$)

Fig. 1: PCA projection and K-Means clustering of log-Mel and MFCC features.

TABLE I: Test Set Performance Across Models and Feature Types

Model	Accuracy	F1-score	Weighted Acc.
Decision Tree (MFCC)	0.6701	0.3658	0.4438
Decision Tree (Log-Mel)	0.6541	0.3536	0.4244
KNN (MFCC)	0.7970	0.3678	0.7190
KNN (Log-Mel)	0.7879	0.3671	0.6868
SVM (MFCC)	0.8294	0.3847	0.8285
SVM (Log-Mel)	0.8221	0.3533	0.8187

Overall, the SVM classifier outperformed both KNN and Decision Tree models in all metrics, with MFCC features yielding the best test performance (82.9% accuracy and 82.8% weighted accuracy). KNN achieved moderately strong results, while the Decision Tree provided lower performance but greater interpretability.

To further investigate model behavior, we examined the confusion matrices:

TABLE II: Confusion Matrix for Decision Tree (Log-Mel)

True \ Pred	Absent	Present	Unknown
Absent	1421	359	91
Present	238	83	21
Unknown	75	15	7

TABLE III: Confusion Matrix for Decision Tree (MFCC)

True \ Pred	Absent	Present	Unknown
Absent	1450	324	97
Present	222	91	29
Unknown	64	26	7

TABLE IV: Confusion Matrix for KNN (Log-Mel)

True \ Pred	Absent	Present	Unknown
Absent	1769	92	10
Present	292	50	0
Unknown	85	11	1

TABLE V: Confusion Matrix for KNN (MFCC)

True \ Pred	Absent	Present	Unknown
Absent	1798	67	6
Present	301	41	0
Unknown	86	9	2

TABLE VI: Confusion Matrix for SVM (Log-Mel)

True \ Pred	Absent	Present	Unknown
Absent	1869	2	0
Present	312	30	0
Unknown	90	7	0

TABLE VII: Confusion Matrix for SVM (MFCC)

True \ Pred	Absent	Present	Unknown
Absent	1866	5	0
Present	292	50	0
Unknown	91	6	0

These matrices show that the `Present` (murmur) class was the most difficult to classify correctly, typically suffering from low recall. The `Unknown` class also had poor performance, reflecting label ambiguity and its limited size. Most models performed well on the `Absent` class, which dominated the dataset.

These observations underline the challenge of detecting rare and uncertain cases and highlight the need for class-sensitive metrics like macro-F1 and weighted accuracy.

VII. MODEL INTERPRETABILITY

The Decision Tree classifier was chosen for feature importance analysis due to its transparency. We examined the most influential features for both MFCC and log-Mel representations.

For MFCC features, the most important coefficients included MFCC₁, MFCC₂₁, and MFCC₀. These coefficients describe spectral envelope variations¹ that are likely informative for murmur detection.

For log-Mel features, the most influential components were Mel₂₄, Mel₀, and Mel₁₉, suggesting that the model emphasizes particular frequency bands when classifying.

¹The spectral envelope refers to the smooth contour that outlines the peaks of the frequency spectrum. It captures how energy is distributed across frequencies and helps distinguish different sound characteristics.

These insights are illustrated in Figure 2, which shows the top 10 features based on Gini importance² from the trained Decision Trees.

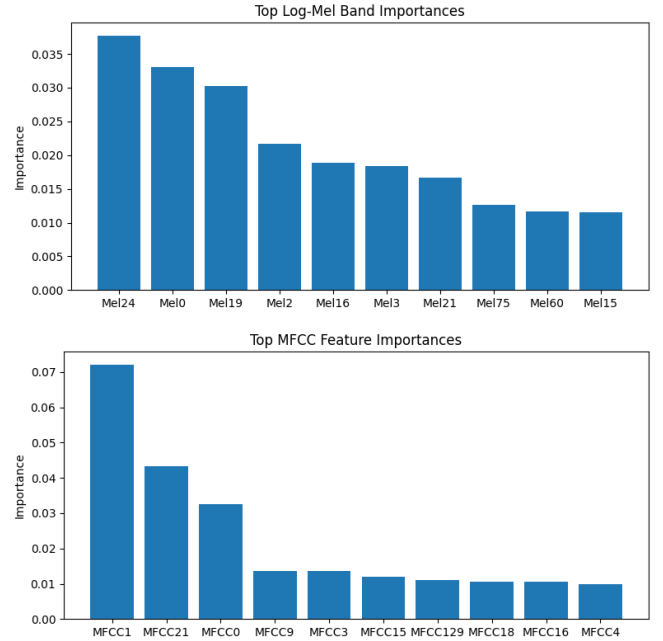


Fig. 2: Top 10 most important features for the Decision Tree classifier. Top: Log-Mel. Bottom: MFCC.

VIII. DISCUSSION

Our results show that both MFCC and log-Mel features can be used to classify heart murmurs effectively using simple machine learning models. Among all models, SVM gave the best performance, followed by KNN and then Decision Tree. MFCC features slightly outperformed log-Mel features, possibly because they are more compact and better capture patterns in heart sound signals.

Although the Decision Tree model had the lowest accuracy and F1-score, it allowed us to understand which features influenced decisions, which is helpful in medical settings. KNN worked well with little setup, and SVM with MFCCs reached the highest performance: 82.9% accuracy, 82.8% weighted accuracy, and 38.7% macro-F1.

To compare with the deep learning approach of Zhang et al. [2], we also calculated the macro-F1 score, which averages the F1 scores for all three classes (Absent, Present, Unknown) equally:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1}_c$$

Unlike the weighted F1-score, macro-F1 is not affected by class imbalance and is more suitable for judging performance on minority classes.

²Gini importance, also called mean decrease in impurity, measures how often a feature is used to split the data and how much those splits improve class separation in the tree.

Zhang et al. reported a macro-F1 of 65.1% and weighted accuracy of 79.8% using a deep neural network with attention over time and recording location. As shown in Table VIII, although our models had lower macro-F1, our best model achieved slightly higher weighted accuracy. This indicates strong performance on the majority class (*Absent*), but also highlights the difficulty our models faced in detecting rare and ambiguous cases.

TABLE VIII: Comparison with Zhang et al. (2024)

Method	Weighted Accuracy (%)	Macro-F1 (%)
Ours (SVM + MFCC)	82.8	38.7
Ours (SVM + Log-Mel)	81.9	35.3
Ours (KNN + MFCC)	71.9	37.0
Ours (KNN + Log-Mel)	68.7	36.7
Ours (DT + MFCC)	44.4	36.3
Ours (DT + Log-Mel)	42.4	35.3
Zhang et al. (CNN)	78.0	61.9
Zhang et al. (RNN-HSMM)	77.6	62.3
Zhang et al. (HMS-Net)	77.6	64.7
Zhang et al. (Proposed)	79.8	65.1

Although our models are simpler and do not capture complex patterns as deep learning models do, they offer key benefits: they are fast, transparent, and easy to deploy. This makes them practical for real-time use or in clinics where computational resources and explainability are important.

IX. CONCLUSION

In this project, we explored how classical machine learning models can be used to classify heart murmurs from PCG recordings using simple, interpretable features. We extracted MFCC and log-Mel representations and tested them with Decision Tree, KNN, and SVM classifiers.

Our experiments showed that SVM with MFCC features achieved the best overall results. While deep learning models like Zhang et al.'s achieved higher macro-F1 scores, our approach had slightly better weighted accuracy and was easier to interpret and implement.

REFERENCES

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- [3] George B. Moody PhysioNet Challenge 2022: Heart Murmur Detection from Phonocardiogram Recordings. Available: <https://physionetchallenges.org/2022/>